Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS7CM05]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Explain the PEAS description for medical diagnosis system and satellite image analysis system.	**07**
	b.	Write short notes on fully observable and deterministic task environments.	**07**
2.	**a.**	Explain breadth first search with example.	**07**
	b.	Explain Djiktra's algorithm with example.	**07**
3.	**a.**	What do you mean by terms? Explain with examples.	**07**
	b.	Write short notes on atomic and complex sentences in first order logic.	**07**
		OR	
4.	**a.**	Explain nested quantifiers with examples.	**07**
	b.	Explain the connections between universal and existential quantifiers.	**07**
5.	**a.**	Write short notes on variance and standard error.	**07**
	b.	Explain conditional log likelihood and mean squared error.	**07**
		OR	
6.	**a.**	Explain learning decision trees from examples.	**07**
	b.	Write short notes on model selection in machine learning.	**07**
7.	**a.**	Write short notes on Simulating Basic Machine Learning with Shallow Models and Radial Basis Function Networks.	**07**
	b.	Explain recurrent neural networks.	**07**
		OR	
8.	**a.**	Discuss briefly the basic architecture of perceptron.	**07**
	b.	Write short notes on multilayer neural network.	**07**